

NAZIRU HALILU

AI · Computer Vision · Geospatial ML for Digital Agriculture

Pamplona, Spain | +34 6 5296 7216 | halilu.175366@e.unavarra.es

[GitHub](#) | [ORCID](#) | [Zenodo records](#) | [Portfolio](#)

Agricultural engineer with a long-term research focus on the digitalization of agriculture, spanning computer vision, geospatial machine learning, agricultural control systems, and high-throughput phenotyping, grounded in a strong foundation in agricultural machinery, mechanization, and farm power systems. Current interests include self-supervised deep learning for crop-stress detection and vegetation-index estimation from UAV and multispectral imagery, and closed-loop sensing and control for irrigation, greenhouse automation, and precision farm machinery.

This document is a complete, single-file capture of every section of the live portfolio site (halilunaziru73-creator.github.io), including all gallery photographs and certificate images, generated for offline reading and archival purposes.

EDUCATION

- Erasmus Mundus Joint Master's Degree (EMJMD), Cutting-Edge Technologies for Sustainable Agriculture** 2025 to expected 2027
Public University of Navarre, Spain · UTAD, Portugal · Agricultural University of Athens, Greece
- Bachelor of Engineering (B.Eng.), Agricultural and Bioresources Engineering, First Class Honours** 2019 to 2024
Ahmadu Bello University, Zaria, Nigeria · CGPA 4.77/5.00, top 1% of graduating cohort
- National Diploma, Agricultural Engineering Technology, Distinction** 2016 to 2018
Ahmadu Bello University, Zaria, Nigeria · CGPA 3.78/4.00, top 1% of graduating cohort

RESEARCH EXPERIENCE

- EMJMD Research Scholar (Mobility Track)** Sep 2026 to Jan 2027
Agricultural University of Athens, Greece
- Engineered NaCROP, a modular Python irrigation and crop-growth decision-support pipeline with a Tkinter GUI, validated across five smallholder crops around Zaria, Nigeria. [Code: GitHub](#)
- Summer Intern, Computer Vision Group** Jun 2026 to Aug 2026
Dept. of Computer Engineering, Ahmadu Bello University, Zaria
- Extended the GACL framework with four new geometry-invariant components, reaching 64.4% accuracy (macro AUC 0.882) across 19 disease classes. [Code: GitHub](#)
- EMJMD Research Scholar (Mobility Track)** Feb 2026 to Jun 2026
University of Trás-os-Montes and Alto Douro (UTAD), Portugal
- Integrated GIS, remote sensing, and ML pipelines across three cropping systems (R-squared 0.978–0.995), cutting lime demand by 2.69 tonnes. [Code: GitHub](#)
 - Developed Multitask Singularity Regression Kriging (MSRK), reducing residual kriging uncertainty by roughly 20% across 17,864 soil samples. [Code: GitHub](#)
 - Built an NDVI/logarithmic regression model forecasting grape ripening (R-squared = 0.9784) across a 6-hectare UTAD vineyard.
 - Ran terrain characterisation and a 25-trap spatial monitoring network for grape moth across a 42.97-hectare Douro vineyard estate.
- EMJMD Research Scholar (Mobility Track)** Sep 2025 to Jan 2026
Public University of Navarre, Spain
- Built a 14-module open-source pipeline for RGB/multispectral vegetation indexing, GLCM texture analysis, and CNN/Faster R-CNN inference, reaching 33.8% balanced accuracy against a 4.55% chance baseline. [Code: GitHub](#)
 - Designed STADOS, a five-stage radiometric harmonization framework, cutting temporal NDVI variance by 39% and 31% on Sentinel-2 imagery. [Code: GitHub](#)
 - Built a Bayesian digital twin combining hydrological, hydraulic, and hybrid CNN-BiLSTM-attention sediment models for gully erosion biocontrol (hold-out R-squared = 0.90). [Code: GitHub](#)
- Undergraduate Project Assistantships** Mar 2024 to Mar 2025
Depts. of Agricultural and Bioresources Engineering and Animal Science, ABU
- Contributed data collection, statistical analysis, and literature review across concurrent undergraduate research projects spanning irrigation, soil erosion, postharvest, and animal systems.
- Project Assistant (Remote Data Lead), Smart Greenhouse Automation** 2024 to 2025
IAR-ABU, Nigeria
- Automated greenhouse environmental control by engineering adaptive irrigation-scheduling algorithms in Python.
- Student Research Assistant, Precision Crop Systems** 2022 to 2024
NAERLS, Nigeria
- Trained CNN-based crop-stress detection models in TensorFlow, reaching 78% classification accuracy on multispectral UAV imagery.
- Assistant Farm Manager (concurrent, part-time)** 2022 to 2024
Yankari Nasarawa Farms, Nigeria

- Managed automated climate-control and water-dosing systems across a commercial poultry facility.

Project Assistant, Smart Irrigation and Water Systems

2021 to 2022

BOMO-Basin Project, Nigeria

- Automated water delivery across a one-hectare smart drip-irrigation array via closed-loop control logic.

Research Intern

2017 to 2020

NCAM, Ilorin and IAR, Zaria, Nigeria

- Built Python data-logging pipelines benchmarking mechanization efficiency; supported baseline hydrological research.

Founder, Nhalilu's Biogas Plant and Sales Company

Registered with SMEDAN, Bioprocess Engineering

- Founded and registered a biogas production and sales venture applying bioprocess engineering principles.

PUBLICATIONS

Peer-Reviewed

- **Halilu, N.** (2026). A Hybrid CNN-BiLSTM-Attention Deep Learning Framework for Sediment Transport in an Agricultural Gully System. *IJRSI*, 13(7), 2920–2950. [DOI](#) | [Code](#)
- **Halilu, N.,** Sulaiman, J. (2026). Real-Time RGB Proxy Vegetation Indexing and Texture Analysis for UAV and Handheld Crop Imagery. *IJRSI*, 13(7), 2951–2989. [DOI](#) | [Code](#)
- **Halilu, N.,** Sulaiman, J. (2026). Sensing the Invisible: A Critical Review of NIR and Hyperspectral Analysis for Crop and Food Quality Assessment. *IJRSI*, 13(7), 2990–3020. [DOI](#)
- Ismail, K., **Halilu, N.** (2024). AI-Driven Precision Agriculture System for Optimizing Maize Yield in Northern Nigeria. *Proc. AfCPA*, 2(1), 100–120.
- Auwal, M., Yusuf, A., **Halilu, N.** (2020). Physio-chemical and bacteriological quality of water for irrigation purpose (Shika Dam). *Nigerian Research Journal of Engineering and Environmental Sciences*, 5(1), 198–204.

Submitted / Under Review

- **Halilu, N.,** Sulaiman, J. Geometry-Agnostic Contrastive Learning (GACL). *Biosystems Engineering* (Elsevier). Under review. [Code](#)
- **Halilu, N.** (2026). Digital Twin for the Evaluation of Experimental Gully Biocontrol Using Morning Glory. *Environmental Monitoring and Assessment*. Under review. [Code](#)
- **Halilu, N.,** Sulaiman, J. Operationalizing GIS and Machine Learning across Contrasting Cropping Systems. *European Journal of Agronomy*. Under review. [Code](#)

Preprints

- **Halilu, N.** (2026). Multitask Singularity Regression Kriging. *Zenodo Preprint*. [Code](#)
- **Halilu, N.** (2026). STADOS: Spectral-Temporal Adaptive Dark Object Subtraction. *Zenodo*. [Code](#)
- Halilu, N. (2025). A Systematic Review of IoT, Edge Computing, Time-Series Modelling and Control Systems in Precision Irrigation. *Zenodo*.

RESEARCH PROJECTS AND OPEN-SOURCE REPOSITORIES

- Geometry-Agnostic Contrastive Learning (GACL): <https://github.com/halilunaziru73-creator/Geometry-Agnostic-Contrastive-Learning-GACL>
- STADOS: Spectral-Temporal Adaptive Dark Object Subtraction: <https://github.com/halilunaziru73-creator/STADOS-Spectral-Temporal-Adaptive-Dark-Object-Subtraction>
- Multitask Singularity Regression Kriging (MSRK): <https://github.com/halilunaziru73-creator/Multitask-Singularity-Regression-Kriging-for-Spatio-Temporal-Soil-NPK-Crop-Stress-Yield-Prediction>
- Digital Twin for Gully Biocontrol Using Morning Glory: <https://github.com/halilunaziru73-creator/Digital-Twin-for-the-Evaluation-of-Experimental-Gully-Biocontrol-Using-Morning-Glory-Ipomoea-spp>
- Hybrid CNN-BiLSTM-Attention Sediment Transport Framework: <https://github.com/halilunaziru73-creator/Hybrid-CNN-BiLSTM-Attention-Sediment-Transport-Agricultural-Gully-System>
- Real-Time RGB Proxy Vegetation Indexing and Texture Analysis: <https://github.com/halilunaziru73-creator/Real-Time-RGB-Proxy-Vegetation-Indexing-and-Texture-Analysis-for-UAV-and-Handheld-Crop-Imagery>
- Operationalizing GIS and ML across Contrasting Cropping Systems: <https://github.com/halilunaziru73-creator/Operationalizing-GIS-and-Machine-Learning-across-Contrasting-Cropping-Systems>
- Geospatial Data Analysis (six-notebook GIS course): <https://github.com/halilunaziru73-creator/Geospatial-data-analysis>
- GIS-Based Livestock Slurry Suitability Delineation: <https://github.com/halilunaziru73-creator/GIS-based-delineation-of-areas-suitable-for-livestock-slurry-application>
- GIS Monitoring of Grape Moth (*Lobesia botrana*): <https://github.com/halilunaziru73-creator/GIS-Monitoring-of-Grape-Moth-Vineyard-Pest>

- GIS Monitoring of Vineyard Ripening (Brix): <https://github.com/halilunaziru73-creator/GIS-Monitoring-of-Vineyard-Ripening-Brix>
- NaCROP: Crop ET and Irrigation Decision-Support Pipeline: <https://github.com/halilunaziru73-creator/NaCROP>
- Integrated Decision-Support Model for Dairy Processing: <https://github.com/halilunaziru73-creator/Integrated-Decision-Support-Model-for-Dairy-Processing-under-Milk-and-Market-Variability>
- Full repository index: <https://github.com/halilunaziru73-creator>

SKILLS

Geospatial AI / GIS: QGIS, ArcGIS, ENVI, Google Earth Engine, Sentinel imagery, UAV imaging

Machine Learning: PyTorch, TensorFlow, Keras, scikit-learn, CNNs, self-supervised learning

Crop Modelling: APSIM Next Generation, crop growth simulation

Climate AI: CMIP6, ERA5, ACCESS-S2, downscaling and forecasting

3D Computer Vision: COLMAP, Open3D, Pix4D, Agisoft Metashape

Programming: Python, MATLAB, R, Git/GitHub, Zenodo

TEACHING, LEADERSHIP AND COMMUNITY

Undergraduate Course Tutor

2020 to 2024

ABU Zaria (NAMET)

- Ran supplementary tutorials in Differential Equations and Transforms, Irrigation System Design, and Computer Applications in Agricultural Engineering.

Secretary General, NAAESB

2023 to 2024

National Assoc. of Agric. Engineering Students

- Directed operational targets and inter-chapter scientific meetups for the national student body.

Deputy Coordinator

2020 to 2025

KOJYAF, Zango Shanu, Zaria

- Coordinated tutorial and mentorship programmes preparing secondary students for JAMB entrance examinations.

Director, Environmental Protection Agency

2023 to 2024

ABU Students' Union Government

- Directed campus environmental protection and sanitation initiatives.

Physics and Mathematics Teacher

2022 to 2023

Takai Academy, Zaria, Nigeria

- Instructed senior classes in classical physics and computational principles.

AWARDS AND HONOURS

- Erasmus Mundus Joint Master Scholarship, European Union (2025)
- Best Project Award, ABU Senate (2024)
- Best Graduating Student, Agricultural and Bioresources Engineering (2024)
- Nigerian Federal Government Award for Academic Excellence (2024)
- Best Undergraduate Research Paper Award, ABU (2024)
- Best 300 Level Student, ABU (2022)
- Best 200 Level Student, ABU (2021)
- Award of Merit as Tutor, Faculty of Engineering, ABU (2021)
- Best Graduating Student, National Diploma, Samaru College of Agriculture (2019)
- PTDF Undergraduate Fellow (2023)
- Needs-Based University Fellowship (2022)
- Award of Honour, International Career Coaching Program, Envirofly Consulting under UK FCDO (2024)

- Award for Best Graduating Student, National Union for Engineering Student Association (2024)
- Award of Merit, Al-Azhar Institute of Arabic and Islamic Studies (2022)
- Award of Merit as Senator, Student Union Government, Samaru College of Agriculture (2017)
- Best Student in Science Subjects, Government Secondary School Jama'a (2014)

CONFERENCES, SEMINARS AND WORKSHOPS

- Data Management Workshop: From Phenotyping Experiment to FAIR Data, Norway (2026)
- Digital Agroecology Cluster Conference (EU-funded), Brussels (2025)
- 6th African Regional Conference on Irrigation and Drainage, ARCID (2025)
- Vibrational Spectroscopy and Chemometrics in Agro-Food Systems (EU-funded) (2025)
- Mechanistic Modeling of Rainfall-Driven Soil Solute Transport, University of Florida (2025)
- IEEE Intl. Conf. on Smart Agriculture Technologies (AgriTech), Valencia, Spain (2024)
- NAERLS Technical Seminar Series: AI in the Food Value Chain (2023)

PROFESSIONAL MEMBERSHIPS

American Society of Agricultural and Biological Engineers (ASABE) · Nigerian Institution of Agricultural Engineers (NIAE) · Nigerian Society of Engineers (NSE) · Pupil Engineer, COREN

REFEREES

Prof. Javier Casalí Sarasibar, Public University of Navarre, Spain · jcs@unavarra.es

Prof. Konstantinos Soulis, Agricultural University of Athens, Greece · soco@aua.gr

Prof. Miguel Ángel Campo Bescós, Public University of Navarre, Spain · miguel.campo@unavarra.es

Prof. José Tadeu Marques Aranha, UTAD, Portugal · j_aranha@utad.pt

Dr. Ibrahim Babangida Dalha, Ahmadu Bello University, Nigeria · ibdalha@abu.edu.ng

Dr. Habibu Ismail, Ahmadu Bello University, Nigeria · habfta@yahoo.com

Dr. Mohammed Abdulsalam, Ahmadu Bello University, Nigeria · mabdulsalam@abu.edu.ng

Dr. Yahya Gambo, KFUPM, Saudi Arabia · yahya.Gambo@kfupm.edu.sa

CERTIFICATIONS AND DOCUMENTED RECOGNITIONS

Honorary Awards



Award of Merit, Senator (SRC/SUG)

Samaru College of Agriculture SUG, 2017/2018 session



Certificate of Merit, Best 200 Level Student

Dept. of Agric. & Bio-Resources Engineering, ABU, 2018/2019 session



Best Overall Graduating Student, National Diploma

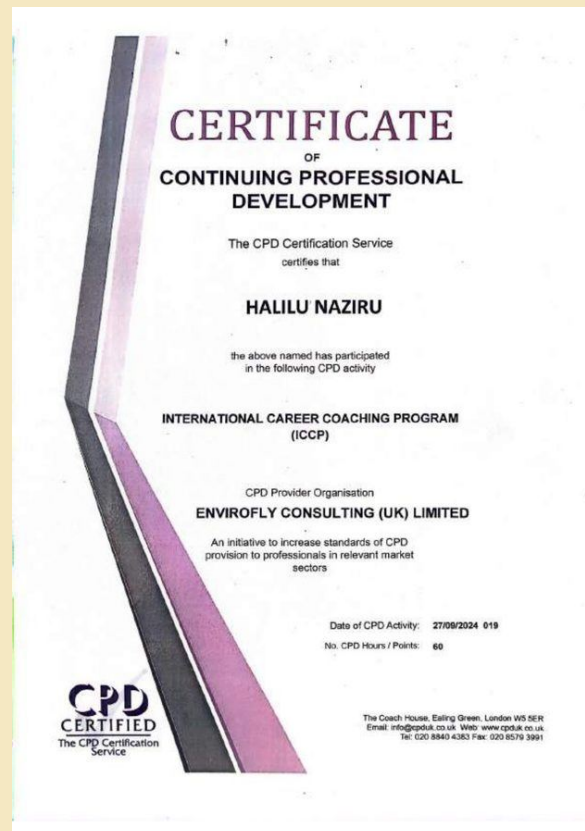
Samaru College of Agriculture, ABU, 2017/2018 session



Best Graduating Student, ND Agric. Engineering Technology

Samaru College of Agriculture, ABU, 2017/2018 session

Training



Certificate of Continuing Professional Development

ICCP, Envirofly Consulting (UK), 60 CPD hours, 2024



Tony Elumelu Foundation Certificate

Business Management Training, 2020



Certificate of Attendance, Explore AI in Bioinformatics

Genomac Institute Inc., USA, 2025



Certificate of Learning, Genomics and Bioinformatics Bootcamp

Genomac Institute Inc., USA, 2024



International Webinar Series for Life Scientists

Genomac Institute Inc., USA, 2024



Bioinformatics Course 201

Genomac Institute Inc., USA, 2024



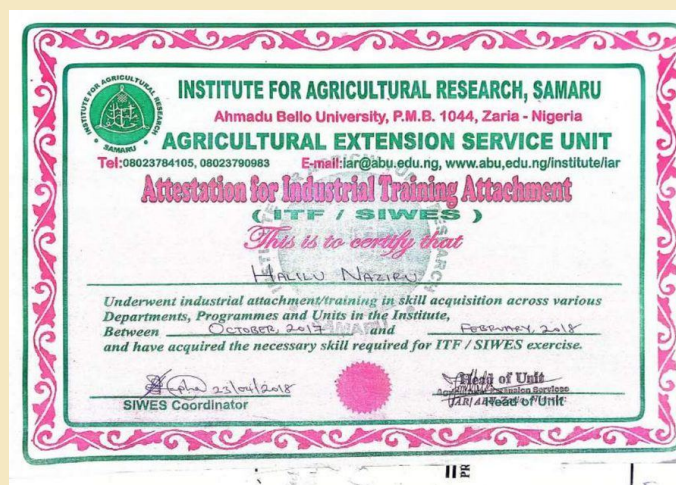
Genomics and Bioinformatics Short Course 301

Genomac Institute Inc., USA, 2024



SIWES Industrial Attachment Attestation

NAERLS, ABU, 2023



SIWES Industrial Attachment Attestation

IAR, Samaru, ABU, 2017/2018

IDOYA LABAYEN GOÑI, directora de ISFOOD Instituto de Innovación y Sostenibilidad en la Cadena Agroalimentaria de la Universidad Pública de Navarra,

HACE CONSTAR

Que **NAZIRU HALILU** ha participado como asistente en el evento "Mechanistic modeling of the impact of rainfall pumping on soil solute remobilization into runoff", por Rafael Muñoz-Carpena - University of Florida (EEUU) * celebrado el 7/11/2025 Sala de conferencias del edificio Jerónimo de Ayaz (Campus de Arrosadía) desde la Universidad Pública de Navarra, con una duración de 12:30h a 13:30h.

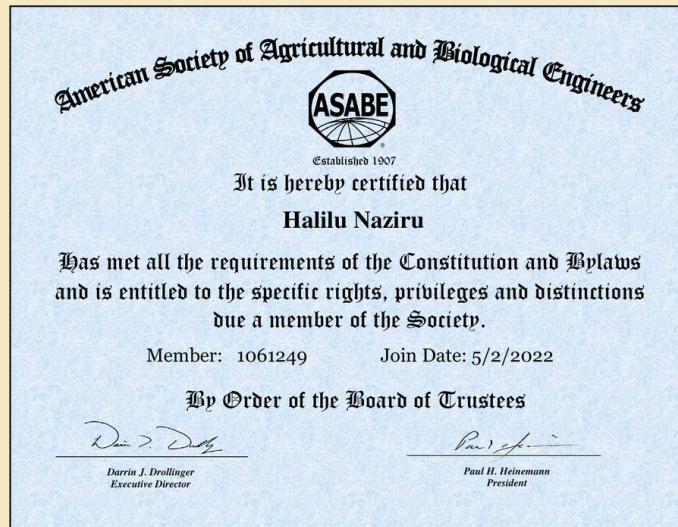
Pamplona, 7/11/2025

Idoya Labayen Goñi
Directora ISFOOD

Mechanistic Modeling Seminar Attendance

ISFOOD Institute, UPNA, 2025

Recognition



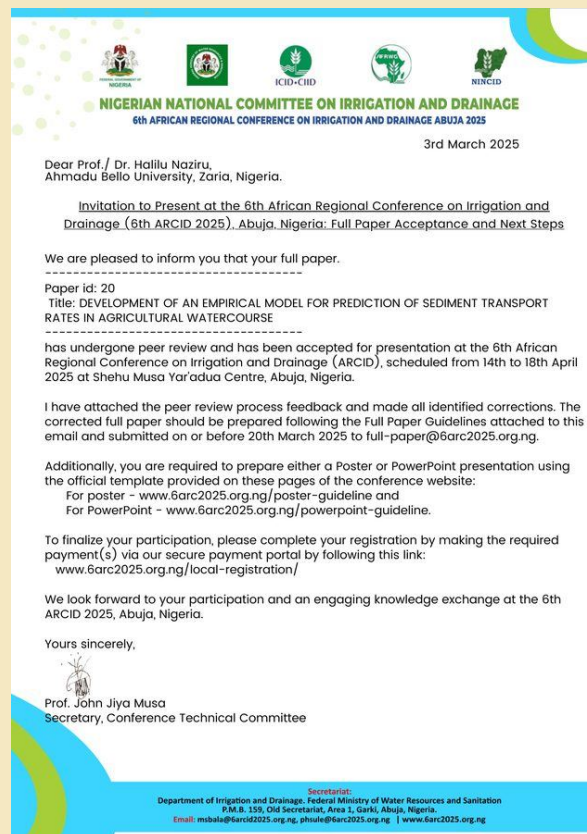
ASABE Member Certificate

Member No. 1061249, joined 2022



NIAE Student Member Certificate

Membership No. S2644, 2024



Conference Paper Acceptance Letter

6th ARCID 2025, Abuja



IJRSI Publication Certificate

Sediment Transport paper, Vol. 13 Issue 7, 2026



IJRSI Publication Certificate

Real-Time RGB Vegetation Indexing paper, Vol. 13 Issue 7, 2026

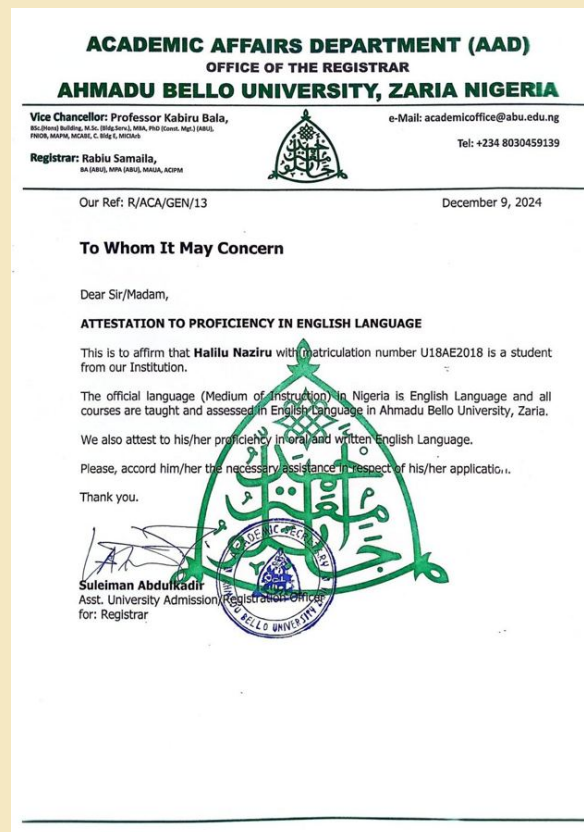


IJRSI Publication Certificate

NIR/Hyperspectral review paper, Vol. 13 Issue 7, 2026



SMEDAN Certificate of Registration
N.halilus Biogas Plant & Sells, 2020



Attestation to Proficiency in English Language
Ahmadu Bello University, 2024

West African Examinations Council

The West African Examinations Council
West African Senior School Certificate

JUNE 2015

This is to Certify that:
HALILU NAZIRU

born on: APRIL 12, 1998

sex: MALE

whose photograph is embossed, having been in attendance at:
KNOWLEDGE IS POWER SECONDARY SCHOOL, ZARIA

sat the West African Senior School Certificate Examination
and obtained the results shown below:

SUBJECT	GRADE
GEOGRAPHY	A1
CIVIC EDUCATION	B3
ENGLISH LANGUAGE	C6
MATHEMATICS	A1
AGRICULTURAL SCIENCE	B3
BIOLOGY	C6
CHEMISTRY	C4
PHYSICS	C5
COMPUTER STUDIES	B2
SUBJECT RECORDED	NINE

CD 15

Candidate No.
4191930088

Certificate No.
**NGWASSCS
21236230**


Chairman of Council


Registrar to Council

Any alteration, erasure or absence of photograph renders this Certificate invalid.

West African Senior School Certificate (WAEC)

Knowledge Is Power Secondary School, Zaria, 2015

FULL ACADEMIC RECORD

Course-by-course record across National Diploma, B.Eng., and EMJMD programmes. Full authenticated transcripts are legalized by the Federal Ministry of Education, Nigeria, and the Consulado General de España en Lagos.

ND Grading Scale

Score %	Letter Grade (LG)	Weighted Points (WP)	CGPA/Grade of Certificate	
80 – 100	A	4	3.5 – 4.0 Distinction	A
70 – 79	AB	3.5	3.0 – 3.49 Upper Credit	B
60 – 69	B	3	2.5 – 2.99 Lower Credit	C
50 – 59	BC	2.5	2.0 – 2.49 Pass	D
40 – 49	C	2	Below 2.0 Fail	F
30 – 39	CD	1.5		
20 – 29	D	1		
10 – 19	E	0.5		
0 – 9	F	0		

CGPA Achieved: 3.78/4.00, Distinction

B.Eng Grading Scale

Note

1) CLASSIFIED DEGREE

SCORE (%)	LETTER GRADE	GRADE POINT	CGPA	CLASS OF DEGREE
70 – 100	A	5	4.50 - 5.00	1 st Class
60 – 69	B	4	3.50 - 4.49	2 nd Class Upper
50 – 49	C	3	2.40 - 3.49	2 nd Class Lower
45 – 49	D	2	1.50 - 2.39	Third Class
40 – 44	E	1	1.00 - 1.49	Pass
0-39	F	0	.00 - .99	Fail

2) NON-CLASSIFIED DEGREE

(To be inserted by the Faculty concerned)

CONSULADO GENERAL DE ESPAÑA EN LAGOS

Visto Bueno para legalizar la firma y/o el sello de KINGSLEY ADIABEKPA

Ministerio de Asuntos Exteriores de Nigeria por ser, al

CONSULADO GENERAL DE ESPAÑA EN LAGOS
Visto Bueno para legalizar la firma y/o el sello de
Ministerio de Asuntos Exteriores de Nigeria por ser, al

CGPA Achieved: 4.77/5.00, First Class

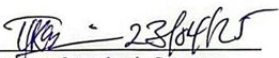
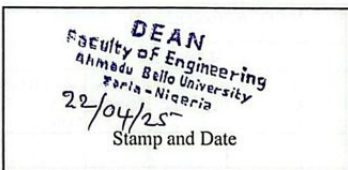
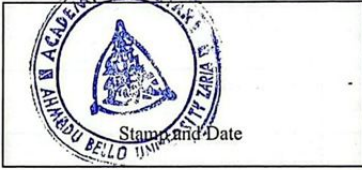
Signatures and Official Seals

Head of Department, Prof. Henry Igbadun

MMENS23	Welding And Brazing	2	A	5	10
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Dept. of Agric & Bio-Resource Engineering, ABU, 2025/2026 session

Dean, Engr. Prof. Jibril Muhammad Kaura & Academic Secretary

26/03/2025 25131010005259 69900-00 NGR CAJERO	Total Credit Units (Registered)	ABU Cum Credit Units (Earned)	ABU CGPA
			
Signature of Dean		Signature of Academic Secretary	
			

Faculty of Engineering, Ahmadu Bello University, dated 22/04/25

Federal Ministry of Education & El Cónsul General, José María Ferré Trad



Federal Government of Nigeria & Consulado General de España en Lagos, dated 26 May 2025

Dr. A. Salisu

	132.0		GPA: 3.89	36		140.0
..... CERTIFICATE AWARDED: National Diploma.....						
SIGNATURE:						
..... GRADE: ...Distinction... DATE: ...9/04/2025... NAME:Dr. A. Salisu.....						

Samaru College of Agriculture, ABU, National Diploma certificate, 2016 to 2018

Course Categories (10 disciplines, full course-by-course marks on the live site)

- Agricultural Machinery, Mechanization & Power (10 courses)
- Soil, Water, Irrigation & Environmental Engineering (18 courses)
- Data Science, Precision Agriculture & Remote Sensing (12 courses)
- Processing, Post-Harvest & Biological Systems (8 courses)
- Mechanical & Materials Engineering (21 courses)
- Electrical, Instrumentation & Structures (10 courses)
- Mathematics, Statistics & Surveying (11 courses)
- Agronomy, Soil Science & Animal Sciences (4 courses)
- Management, Economics & General Studies (28 courses)
- Projects, Practical Exercises & Industrial Training (9 courses)

PLACES WORKED AND STUDIED

From Zaria, Nigeria, through Spain (Pamplona, Madrid, Barcelona), Portugal, Greece, Qatar, Morocco, Ethiopia, France, Italy, and Germany, over the course of research travel, conferences, and the Erasmus Mundus programme.

- Nigeria (home base)
- Spain (EMJMD Term 1, UPNA)
- Portugal (EMJMD Term 2, UTAD)
- Greece (EMJMD Term 3, AUA)
- Qatar (tourism)
- Morocco (tourism)
- Ethiopia (tourism)
- France (conference and tourism)
- Italy (conference and tourism)
- Germany (conference and tourism)

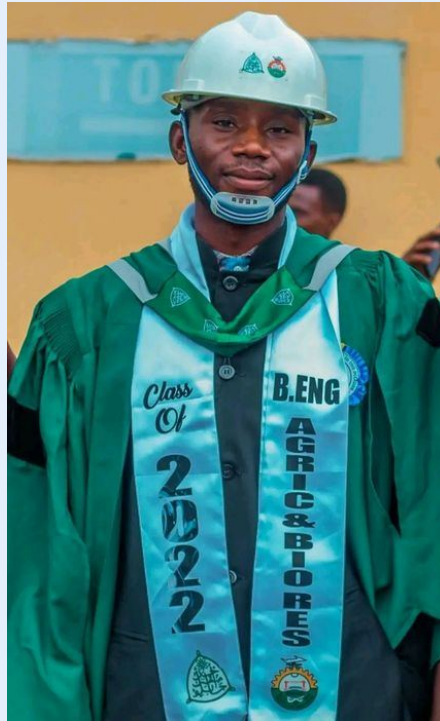
RECENT UPDATES

- Sep 2026: Published GIS monitoring datasets for grape moth and vineyard ripening (Brix), archived on Zenodo with DOIs.
- Aug 2026: Two papers accepted at IJRSI, covering sediment transport modelling and real-time vegetation indexing.
- Aug 2026: Archived STADOS on Zenodo, alongside updated releases of the Zaria irrigation DSS and NaCROP.
- Mar 2025: Paper accepted for presentation at the 6th ARCID, Abuja, on sediment transport rate prediction.

GALLERY

Field and campus photographs, organized by theme.

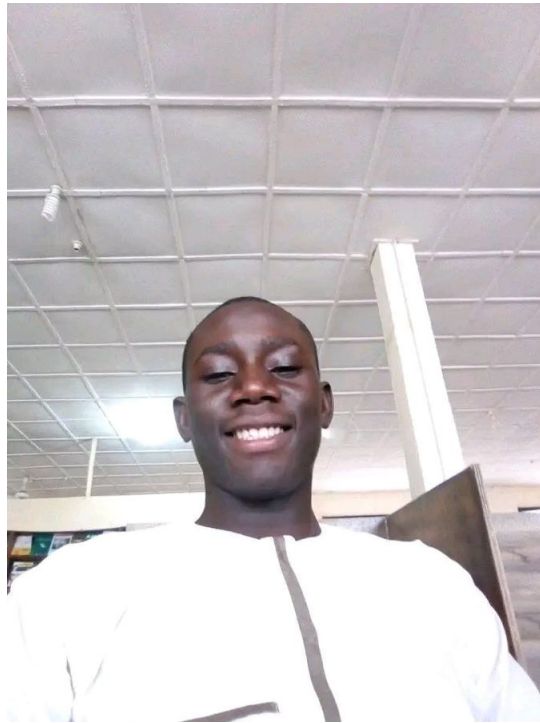
Graduation Ceremony (3)



B.Eng graduation ceremony, Class of 2022 cohort (graduated 2024 after COVID-19 and ASUU strike disruptions), Agric & Bio-Resources Engineering, 2019 to 2024



Graduation ceremony procession, Bachelor of Engineering, 2019 to 2024



National Diploma graduation celebration, Agricultural Engineering Technology, 2016 to 2018

EMJMD Programme & Welcome Events (10)



Large group photo on a stone terrace, EMJMD cohort, 2025



Formal welcome event in a university corridor with staff and students, 2025



Group photo beside a metal sculpture installation, wind turbines in the distance, 2025



Receiving welcome bags on arrival at UTAD, 2026



Wearing an EU Global Gateway shirt for an EMJMD event, 2025



Outside the Agricultural University of Athens, 2026



At the Erasmus+ Pre-Departure Orientation event, 2025



In front of the UTAD banner, 2026



Beside the official SUSTAGRI EMJMD programme banner, 2025

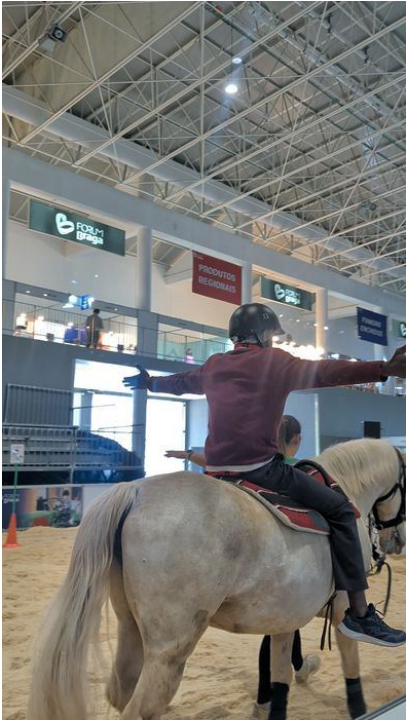


Group photo with the EMJMD SUSTAGRI cohort, 2025

Agricultural Fairs & Machinery Demonstrations (7)



Agricultural trade fair, rice and grain product stands, 2026



Horse-riding demonstration at an indoor arena, Forum Braga, 2026



Dairy trade fair with the ALIP cow mascot, 2026



Farm machinery expo, compact tractor on display, 2026



Farm machinery expo, standing beside tractors, 2026



Trying a John Deere precision-agriculture GPS guidance system, 2026



Braga city tourism display at an expo, 2026

Douro Valley Vineyard Fieldwork (7)



Group visit to Quinta do Seixo vineyard estate, Douro Valley, 2026



Visiting the Quinta S. Luiz port wine estate, Douro Valley, 2026



Irrigation sensor station on the vineyard terraces, 2026



Inside the wine fermentation tank hall, 2026



Walking between rows of stainless steel fermentation tanks, 2026

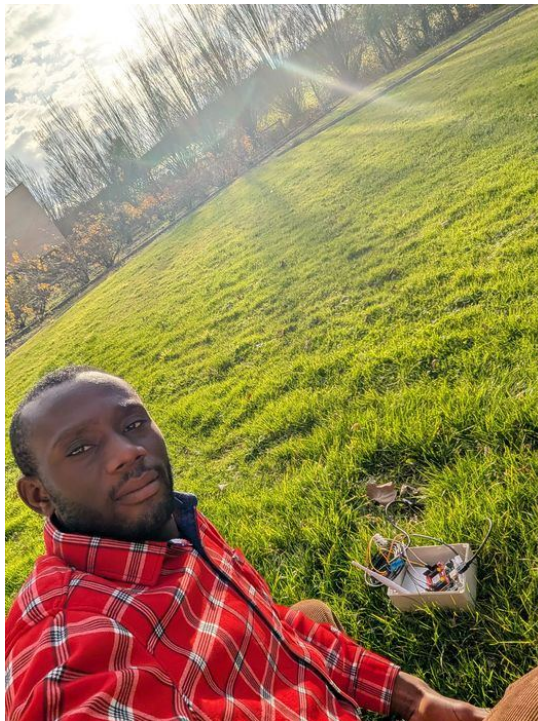


Inspecting a weather station on the vineyard terraces, 2026



Fieldwork at the vineyard weather station, 2026

Sensor & Field Research (3)



Testing a sensor prototype outdoors on the lawn, 2025



Inspecting a LoRa IoT irrigation sensor box in the field, 2025



Hyperspectral imaging fieldwork on apples, snow-covered orchard in winter, 2025

Presentations & Conferences (3)



Classroom presentation on risks and limitations of spectral sensor deployments, 2025



Group selfie with conference lanyards, 2025



Presenting at a conference lectern, 2025

Campus Life & Social Gatherings (11)



Amigo Secreto (Secret Santa) gift exchange in the classroom, 2025



Group celebration after the Secret Santa gift exchange, 2025



Evening gathering with classmates at an apartment party, 2025



Bowling and arcade outing with friends, 2025



Group walk near the Public University of Navarre, 2025



With friends, football jerseys, outdoor gathering, 2024



Casual indoor portrait, 2024



Coffee break with a friend on a balcony, 2026



Casual indoor portrait, 2025



Sharing a meal with a friend in the university cafeteria, 2025



Evening gathering with friends in the kitchen, 2025



Group hike, resting by a mountain stream, 2025



Portrait on an Atlantic coast beach, 2026



Farm building with a solar panel canopy, 2024



Casual indoor portrait, 2024



Standing in a maize field at dusk, 2024



Near an ancient fortress wall in Greece, 2026



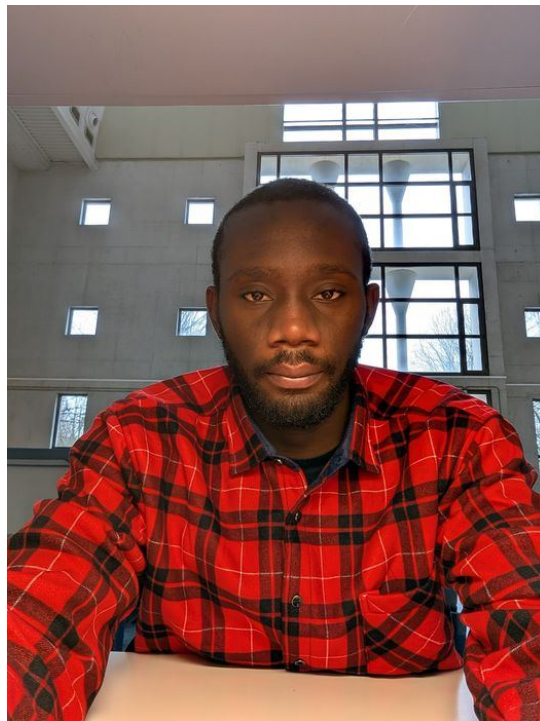
Studying in the university library, 2026



Candid photo with a backpack, purple filter, 2025



Official student portrait for an event, 2025



Studying in a university building, 2026

Greece — Cape Sounion (4)



Cape Sounion coastline, Greece, 2026



Temple of Poseidon ruins, Cape Sounion, Greece, 2026



Sunset over the Aegean Sea, Cape Sounion, Greece, 2026



Temple of Poseidon columns, Cape Sounion, Greece, 2026

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